

Black Hole Problems Answer Key

1. $1.22 \times 10^{10} \text{ m}$

2. $9.76 \times 10^7 \text{ m/s}$

3. $3.68 \times 10^6 \text{ m/s}^2$

4. $2.61 \times 10^{-14} \text{ K}$

5. 0.89 hr , 3196.21 s

6. (a) $\frac{3c^6}{32\pi G^3 M^2}$

(b) $8.15 \times 10^6 \text{ kg/m}^3$